Terahertz Wireless Networks: Opportunities, Challenges, and Enabling Technologies

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***Abstract*—Terahertz (THz) wireless communication offers wide bandwidths for high-data-rate applications, but reliable operation is constrained by propagation loss, molecular absorption, blockage, beam misalignment, and hardware limitations. This narrative review examines these challenges and discusses directional beamforming, antenna arrays, beam management, and adaptive resource allocation as enabling approaches. The review compares selected measurement-based channel-modeling and simulation-based beam-tracking studies, emphasizing their complementary contributions and the limits of their evidence. The analysis highlights that peak link capacity alone is insufficient to assess practical network performance: delivered payload throughput, outage duration, recovery delay, control overhead, and energy consumption must also be considered. The review identifies measurement-informed evaluation, adaptive link management, and experimental validation under realistic mobility and blockage conditions as priorities for developing reliable THz wireless networks.**

***Index Terms*—Terahertz communications, wireless networks, beamforming, channel modeling, network reliability.**

I. Introduction

The demand for high-data-rate wireless services motivates research into frequency bands beyond those commonly used in existing networks. Terahertz (THz) communication is being investigated as one possible approach because of the wide bandwidths available in selected frequency windows [1].

However, wide bandwidth alone does not guarantee a reliable wireless connection. THz network design must account for propagation loss, frequency-dependent molecular absorption, blockage, and hardware limitations. Channel measurements and appropriate models are therefore important for understanding practical deployment conditions [2].

Directional antennas and beamforming can help establish useful communication links. However, narrow beams require alignment between transmitters and receivers. User movement and obstacles can make link maintenance more difficult. These challenges connect physical-layer design with network tasks such as beam management and maintaining connectivity [3].

This article presents a focused narrative review of THz wireless networks. It addresses the following question: How can THz networks benefit from large bandwidth while maintaining reliable connections under propagation, blockage, and beam-alignment constraints?

The review draws on three broad surveys [1]–[3] and examines two original studies on indoor channel modeling and adaptive beam tracking [4], [5]. These studies illustrate complementary approaches to understanding and maintaining THz links. The discussion compares their methods, evidence, and limitations, with particular attention to network reliability. The review is selective rather than exhaustive and does not present new measurements or simulation results.

The review first introduces THz fundamentals and potential applications. It then examines propagation and networking challenges, followed by enabling technologies and a comparison of selected studies. Finally, it discusses design trade-offs, open research directions, and the main conclusions.

II. THz Network Fundamentals

Terahertz (THz) wireless communication generally refers to the use of frequencies between 0.1 and 10 THz [2]. This frequency range offers substantial bandwidth that can support high-capacity wireless links and emerging applications [1]. However, the performance of THz networks depends not only on the available bandwidth but also on the propagation environment, transmission distance, and hardware capabilities.

A. Related Work

Previous studies have examined THz wireless communication from different perspectives. Shafie et al. [1] reviewed major challenges and technological advancements across the physical, link, and network layers, emphasizing spectrum management, antenna technologies, and beamforming.

Han et al. [2] focused on THz channel measurements, propagation characteristics, and channel modeling techniques. Their survey provides an important foundation for understanding the behavior of THz wireless links.

Jiang et al. [3] presented a broader review of THz communications and sensing, covering propagation, transceiver technologies, network architectures, and experimental developments.

Building on these studies, the present review focuses on the relationship between THz propagation characteristics, enabling technologies, and network reliability. Particular attention is given to the trade-offs between communication performance, connection stability, and implementation complexity.

B. Propagation Characteristics

THz signals experience free-space propagation loss and frequency-dependent molecular absorption [2]. In particular, atmospheric molecules, including water vapor, absorb electromagnetic energy at specific frequencies. This creates transmission windows in which signal attenuation is relatively low compared with neighboring frequencies.

For a line-of-sight (LoS) THz link, the combined free-space spreading and molecular absorption loss can be expressed as

$$L(f,d) = \left(\frac{4\pi f d}{c}\right)^2 e^{\kappa(f)d}, \tag{1}$$

where f is the operating frequency in hertz, d is the transmission distance in meters, c is the speed of light in meters per second, and $\kappa(f)$ is the power absorption coefficient in inverse meters. Here, $L(f,d)$ is a dimensionless power-loss factor in linear scale, not in decibels. The expression combines free-space spreading and molecular absorption for a line-of-sight path; antenna gains and additional losses due to blockage or misalignment are not included [2].

The severity of these effects depends on operating frequency, communication distance, and environmental conditions. Therefore, accurate channel modeling is essential for evaluating the feasibility and reliability of THz wireless links.

Chen et al. [4] conducted channel measurements between 130 and 143 GHz in a typical meeting room using directional antennas. The study examined line-of-sight path loss and the temporal and spatial characteristics of multipath propagation. A hybrid model combining ray tracing and statistical methods was developed and evaluated against the measurements. The model showed better agreement with the measured temporal-spatial characteristics than conventional statistical and geometry-based stochastic models.

For this review, the main implication is that realistic propagation models provide a stronger basis for evaluating THz links than free-space loss calculations alone. However, the reported validation concerns a particular indoor environment and frequency range. Its applicability to other deployment conditions therefore requires further validation. Accurate channel modeling should also be distinguished from network reliability evaluation, which must additionally consider mobility, blockage, and link recovery.

C. Directional Communication

To improve the link budget, THz communication systems commonly rely on high-gain directional antennas and beamforming techniques [1]. These technologies concentrate transmitted energy toward the intended receiver, improving the received signal strength.

However, narrow directional beams require accurate alignment between the transmitter and receiver. Mobility, environmental obstacles, and sudden blockage can interrupt communication and require beam reconfiguration.

As illustrated in Fig. 1, physical blockage can interrupt a directional line-of-sight THz communication link.

Consequently, reliable THz network operation requires coordination between physical-layer communication techniques and network-level mechanisms such as beam tracking, link recovery, and connectivity management [3].

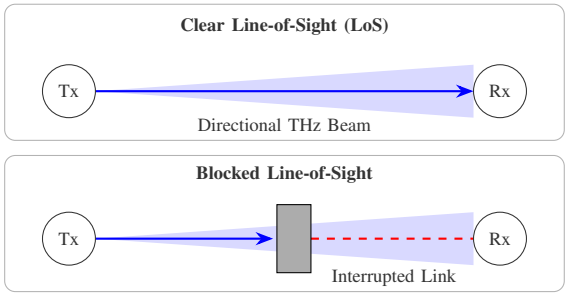


Fig. 1: Conceptual illustration of directional THz communication under clear and blocked line-of-sight conditions.

D. Opportunities and Applications

The availability of large bandwidths in the THz frequency range creates opportunities for extremely high-data-rate wireless communication. Compared with conventional lower-frequency systems, THz technologies can potentially support substantially higher transmission capacities, making them attractive for bandwidth-intensive applications [1].

For an ideal additive white Gaussian noise (AWGN) channel, the theoretical channel capacity is described by Shannon's formula:

$$C = B\log_2(1+\mathrm{SNR}), \tag{2}$$

where C is the theoretical channel capacity in bits per second, B is the available bandwidth in hertz, and SNR is the signal-to-noise ratio in linear scale. This equation illustrates that increasing the usable bandwidth can improve the achievable data rate when the SNR is maintained. However, practical THz communication performance is also limited by propagation loss, hardware constraints, and link reliability [1].

One promising application is short-range, high-capacity indoor communication, where THz links may support rapid data transfer between nearby devices. THz communication is also being investigated for wireless backhaul and fronthaul connections, which can help transport large amounts of data between network infrastructure components. Such applications could reduce reliance on wired connections in suitable deployment scenarios [1].

Another important opportunity is the integration of wireless communication and sensing. The high operating frequencies and large bandwidths of THz systems can support high-resolution environmental sensing and precise localization. Combining communication and sensing capabilities may enable new applications in intelligent transportation, industrial automation, and future wireless networks [3].

Despite these opportunities, their practical realization depends on overcoming propagation, hardware, and networking limitations. Therefore, the potential benefits of THz communication must be evaluated together with the reliability challenges discussed in the following section.

III. Challenges in THz Wireless Networks

While the previous section introduced the fundamental characteristics of THz communication, deploying reliable THz wireless networks presents several practical challenges. These include propagation attenuation, signal blockage, beam misalignment, and hardware limitations [1].

These challenges affect communication range, link stability, and overall network performance. Understanding their impact is essential for evaluating the technologies and design approaches discussed in the following sections.

A. Propagation Loss and Molecular Absorption

One of the major challenges in THz wireless networks is the significant attenuation experienced by signals during propagation. For fixed antenna gains, free-space path loss increases with operating frequency, making the communication link more sensitive to transmission distance. This can reduce the received signal power and limit the achievable communication range [2].

Another important limitation is molecular absorption, which introduces frequency-dependent attenuation. Atmospheric molecules, particularly water vapor, absorb energy at specific THz frequencies. As a result, the available transmission windows and achievable link distances depend on the selected frequency and environmental conditions [2].

From a network design perspective, these propagation characteristics create important trade-offs between bandwidth, coverage, and reliability. Although THz communication offers opportunities for extremely high data rates, practical systems must carefully select operating frequencies, transmission distances, and antenna configurations. Directional beamforming and suitable network deployment strategies can help address these limitations, but they also introduce additional design complexity [1].

B. Blockage and Mobility

Blockage represents a major challenge in THz wireless networks because communication often relies on highly directional line-of-sight (LoS) links. Physical obstacles, such as walls, furniture, and the human body, can significantly attenuate or interrupt the transmitted signal. Although reflected propagation paths may provide alternative connections, these paths can experience additional attenuation and may not always support reliable communication [2].

User mobility introduces further difficulties because the relative positions of transmitters and receivers change over time. Even small movements can affect the alignment of narrow directional beams, potentially reducing the received signal strength and causing temporary link interruptions. Consequently, maintaining stable connectivity requires continuous monitoring of changing channel and network conditions [1].

From a networking perspective, blockage and mobility can increase the frequency of beam realignment, link recovery, and handover procedures. These operations introduce signaling overhead and may increase communication latency, especially in dynamic environments with moving users and obstacles. Therefore, reliable THz networks require efficient mobility management and mechanisms capable of responding rapidly to sudden changes in connectivity [3].

C. Beam Alignment and Tracking

Beam alignment is a critical challenge in THz wireless networks because communication systems typically employ highly directional antennas to achieve sufficient received signal strength. Establishing a reliable connection requires accurate alignment between the transmitter and receiver beams. Misalignment can significantly reduce antenna gain and degrade communication performance [1].

Maintaining beam alignment becomes more difficult in dynamic environments where users or devices change their positions or orientations. Beam tracking mechanisms are therefore required to adjust the transmission direction and maintain connectivity.

However, frequent beam training and realignment procedures introduce additional signaling overhead, energy consumption, and communication delays [3].

An important design trade-off exists between beamwidth and alignment complexity. Narrow beams provide higher directional antenna gain but are more sensitive to positioning errors and mobility. Wider beams simplify alignment and improve tolerance to movement, but generally provide lower directional gain. Consequently, efficient beam management strategies must balance link quality, tracking overhead, and network reliability [1].

D. Hardware and Energy Constraints

The practical implementation of THz wireless networks faces significant hardware challenges due to the extremely high operating frequencies. Generating, transmitting, and detecting THz signals require specialized electronic and photonic components. Current hardware technologies face limitations related to output power, receiver sensitivity, conversion efficiency, and phase noise. These limitations can reduce communication range and affect overall link performance [1].

Energy consumption is another important concern, particularly in systems employing large antenna arrays and advanced beamforming architectures. Signal processing, beam training, and the operation of multiple radio-frequency components can increase power requirements. This creates challenges for mobile devices and energy-constrained network nodes, where battery capacity and thermal management must be considered [3].

From a network design perspective, hardware limitations influence coverage, deployment cost, and energy efficiency. Increasing transmission power or using more complex antenna architectures may improve link performance, but these approaches can also increase hardware complexity and power consumption. Therefore, practical THz networks must balance communication capacity, reliability, hardware feasibility, and energy efficiency [1].

E. Network Reliability

Network reliability is a critical concern in THz wireless communication because maintaining stable connectivity depends on several interconnected factors. Propagation attenuation, molecular absorption, physical blockage, and beam misalignment can significantly affect link availability and communication quality. These challenges become particularly important in dynamic environments where channel conditions change rapidly [1].

Unlike conventional lower-frequency wireless systems, THz networks often depend on highly directional links that can be vulnerable to sudden interruptions. Link failures may cause packet losses, increased latency, and reductions in effective throughput. Therefore, evaluating network performance requires consideration of reliability indicators such as link availability, packet delivery performance, and recovery delay [3].

Improving reliability requires coordinated solutions involving beam management, adaptive transmission techniques, and alternative communication paths. For example, beam realignment and switching to an available backup link may help restore connectivity after blockage. However, these mechanisms introduce additional signaling overhead, processing requirements, and possible delays. Consequently, THz network design must balance high data rates with connection stability, resource efficiency, and recovery performance [1].

IV. Enabling Technologies for THz Wireless Networks

Addressing the challenges discussed in the previous section requires advanced communication technologies and efficient network design strategies. Several enabling approaches have been investigated to improve the capacity, coverage, and reliability of THz wireless networks. These include directional beamforming, large antenna arrays, beam management techniques, and adaptive communication mechanisms [1].

Although these technologies can mitigate propagation limitations and improve link stability, each introduces trade-offs involving hardware complexity, energy consumption, and signaling overhead. Understanding these trade-offs is essential for developing practical and reliable THz communication systems [3].

A. Directional Beamforming

Directional beamforming is one of the key enabling technologies for THz wireless communication. It improves link performance by concentrating electromagnetic energy toward the intended receiver rather than distributing it equally in all directions. This directional gain helps compensate for high propagation losses and supports more reliable high-data-rate communication over suitable distances [1].

Beamforming can be implemented using analog, digital, or hybrid architectures. Analog beamforming controls antenna signals using radio-frequency phase shifters and generally requires fewer RF chains. Digital beamforming provides greater flexibility by processing signals separately across multiple RF chains, but often increases hardware complexity and power consumption. Hybrid beamforming combines analog and digital processing to balance performance, flexibility, and implementation cost [3].

Despite these advantages, directional beamforming introduces important limitations. Narrow beams require accurate alignment between the transmitter and receiver, making communication sensitive to mobility and physical blockage. Furthermore, beam training and adjustment procedures consume network resources and may introduce additional latency. Therefore, efficient beamforming design must balance directional gain, implementation complexity, and connection reliability [1].

B. Large Antenna Arrays and MIMO Technologies

Large antenna arrays are important enabling technologies for THz wireless networks because the short wavelengths at THz frequencies allow many antenna elements to be integrated into relatively compact physical structures. By combining signals across multiple antenna elements, these arrays can provide high directional gain, improving the link budget and supporting high-capacity communication [1].

Multiple-Input Multiple-Output (MIMO) technology uses multiple transmitting and receiving antennas to improve wireless communication performance. In THz systems, MIMO architectures can support beamforming and, under suitable channel conditions, spatial multiplexing to increase data transmission capacity. However, the potential multiplexing gains depend on channel characteristics, antenna configuration, and the availability of sufficiently independent spatial communication paths [3].

Despite their advantages, large antenna arrays and MIMO systems introduce significant implementation challenges. Increasing the number of antenna elements may require more complex signal processing, accurate channel estimation, and efficient hardware architectures. Fully digital implementations can also consume considerable energy due to the required radio-frequency chains. Consequently, practical THz systems often consider hybrid antenna architectures to balance communication performance, hardware complexity, and energy efficiency [1].

C. Beam Management and Tracking

Beam management is an essential technology for maintaining reliable communication in THz wireless networks. Since THz systems commonly rely on highly directional transmission, efficient mechanisms are required to establish, monitor, and maintain suitable communication beams. Beam management generally involves initial beam training, beam selection, tracking, and recovery procedures [1].

Beam tracking enables transmitters and receivers to adjust their beam directions when users move or change their orientation. By updating beam configurations according to changing channel conditions, tracking mechanisms can reduce misalignment and help maintain stable connections. However, frequent beam updates may introduce signaling overhead, processing complexity, and additional energy consumption [3].

Beam recovery is particularly important when communication is interrupted by physical blockage or sudden changes in the propagation environment. In such cases, the network may attempt to establish an alternative beam, use an available reflected propagation path, or switch to another access point. The effectiveness of these approaches depends on the availability of suitable links and the time required to restore connectivity [1].

Karacora et al. [5] proposed event-based beam tracking combined with dynamic beamwidth adaptation for THz communications. Their formulation jointly considers maximizing the expected data rate and minimizing outage probability under transmit-power and overhead constraints. The approach adapts beamwidth to balance directional gain against tolerance to misalignment, while triggering channel estimation according to communication conditions. Simulation results showed improved average rate and reduced outages while maintaining low pilot overhead.

For this review, the key implication is that beam management should be evaluated using both communication performance and the resources required to maintain the link. A highly directional beam is useful only when its gain can be maintained without excessive tracking overhead or interruptions. However, the reported evidence is based on simulations rather than a deployed network. The findings should therefore be interpreted within the evaluated conditions, with experimental validation needed before generalizing the benefits to practical deployments.

Therefore, efficient beam management must balance tracking accuracy, recovery speed, and resource consumption. Reducing unnecessary beam training while maintaining responsiveness to mobility and blockage is an important design objective for reliable THz networks [3].

D. Adaptive Communication and Resource Allocation

Adaptive communication techniques play an important role in improving the performance of THz wireless networks under changing channel conditions. Since THz propagation is affected by transmission distance, molecular absorption, and environmental factors, fixed transmission configurations may not provide satisfactory performance in all scenarios. Adaptive mechanisms can adjust communication parameters according to channel quality and network requirements [1].

Adaptive modulation and coding can improve transmission efficiency by selecting suitable modulation orders and coding rates based on channel conditions. Under favorable conditions, higher-order modulation may support increased data rates, while more robust configurations can help maintain communication quality when the received signal becomes weaker. However, these adjustments require reliable channel information and introduce additional control complexity [3].

Resource allocation is particularly important because THz channels exhibit frequency-dependent propagation characteristics. Different frequency bands may experience different levels of molecular absorption and attenuation. Therefore, frequency selection, bandwidth allocation, and transmit power management should consider the propagation characteristics and requirements of individual communication links [2].

Although adaptive communication and resource allocation can improve network efficiency, their effectiveness depends on accurate channel estimation and timely adaptation. Frequent parameter updates may increase signaling overhead and computational demands. Consequently, practical THz systems must balance achievable throughput, link reliability, adaptation complexity, and energy consumption [1].

Table I compares the two original studies discussed in this review, highlighting their evidence and implications for reliable THz communication.

The studies address complementary aspects of reliable THz communication. Chen et al. [4] provide measurement-based evidence for channel modeling, whereas Karacora et al. [5] evaluate adaptive beam management through simulations. Their results cannot be ranked directly because they use different objectives, methods, and performance measures. Taken together, they motivate evaluating beam-management strategies with measurement-informed channel models, while accounting for both outage performance and tracking overhead.

V. Design Trade-offs and Open Research Directions

The comparison in Table I suggests that reliable THz communication requires both realistic channel models and adaptive link management. These contributions address different parts of the problem: channel modeling supports performance prediction, while beam management responds to changing link conditions. This section discusses how these insights inform network design and performance evaluation.

A. Performance, Reliability, and Complexity Trade-offs

A central distinction is between the data rate supported by an aligned link and the useful throughput delivered over time. For this review, useful throughput refers to successfully delivered payload bits per unit time, including periods spent on beam training and link recovery. A design with a high instantaneous rate may therefore provide limited practical benefit if interruptions or control procedures consume a substantial fraction of operating time.

The study by Karacora et al. [5] illustrates the value of considering rate, outage probability, and tracking overhead together. Its findings support evaluating beam adaptation through multiple metrics rather than directional gain alone. However, comparisons between beam-management methods should use consistent mobility, blockage, and transmit-power conditions so that apparent improvements are not caused by different assumptions.

Channel-model selection is equally important when interpreting such comparisons. Chen et al. [4] showed that a measurement-informed hybrid model can better reproduce the measured channel characteristics than the conventional models considered. This motivates testing whether the benefits of a beam-management strategy remain consistent across different channel models and deployment environments.

Finally, communication gains must be weighed against implementation costs, including processing requirements, RF hardware, and energy consumption [1], [3]. Based on this

Table I: Comparison of Selected THz Studies

Study and Setting	Approach	Main Evidence	Scope and Limitations
Chen et al. [4] Indoor meeting room; 130–143 GHz.	Channel measurements and a hybrid ray-tracing and statistical model.	Closer agreement with measured temporal-spatial characteristics than the conventional models considered.	Validation concerns the measured indoor setting and frequency range; it does not establish network-level reliability.
Karacora et al. [5] Simulation-based THz beam tracking.	Event-based channel estimation and dynamic beamwidth adaptation.	Improved average data rate and reduced outages with low pilot overhead in the evaluated simulations.	Simulation evidence; practical benefits require experimental validation under deployment-specific conditions.

synthesis, a useful evaluation should report payload throughput, outage duration, recovery delay, control overhead, and energy consumption. These measures help distinguish improvements in link capacity from improvements in dependable network service.

B. Open Research Directions in THz Wireless Networks

Despite significant progress in THz communication research, several open questions remain regarding the practical deployment of reliable wireless networks. One important research direction is the development of accurate channel models that account for molecular absorption, blockage, mobility, and different propagation environments. Further measurement-based validation is needed to understand how these factors affect real-world THz communication performance [2].

Another promising research direction involves intelligent beam management and mobility support. Future THz networks require efficient methods for predicting beam misalignment, detecting blockage, and identifying alternative communication paths. Predictive and learning-based approaches may help reduce beam training overhead and improve connection recovery. However, their effectiveness must be evaluated under realistic mobility conditions, taking into account computational complexity, prediction errors, and signaling requirements [1].

Hardware efficiency and practical implementation also remain important research areas. Future developments should investigate energy-efficient transceiver architectures, scalable antenna arrays, and signal processing methods that provide high communication performance without excessive hardware complexity or power consumption. These improvements are particularly important for mobile devices and large-scale network deployments [3].

Finally, future research should investigate joint physical-layer and network-layer optimization. Rather than optimizing data rates, beamforming, or resource allocation independently, THz networks could benefit from coordinated approaches that consider throughput, latency, link availability, and energy efficiency together. Experimental validation and realistic network-level evaluations will be essential for determining which approaches can support dependable THz communication in future wireless systems [1], [3].

VI. Conclusions

This review examined how THz wireless networks can exploit wide bandwidth while maintaining reliable connectivity under propagation, blockage, and beam-alignment constraints. The reviewed evidence suggests that addressing this question requires combining measurement-informed channel models, adaptive beam management, and network mechanisms for maintaining and restoring connectivity.

The comparison of the selected studies highlights complementary contributions: channel measurements support realistic performance prediction, while adaptive beam tracking addresses changing link conditions. However, evidence from a specific indoor measurement campaign or simulation setting does not establish performance across all deployment scenarios.

The main conclusion is that peak link capacity alone is insufficient to assess the practical value of THz networks. Evaluation should also consider delivered payload throughput, outage duration, recovery delay, control overhead, and energy consumption. Future work should validate these trade-offs through realistic mobility and blockage scenarios and experimental network evaluations.

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